



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.EE.4

Equivalent Expressions

Lesson 6-6 • Teacher Copy

Name: Type your name

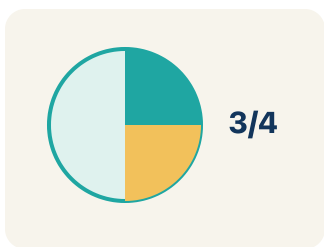
Date: Today's date

Class: Class period

Key Vocabulary

Level 2 Standard

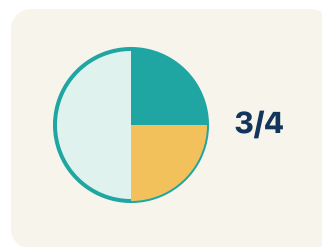
Picture first, then the word, then a plain-language meaning. Say each word out loud.



$2x + 4$ and $2(x + 2)$: when $x = 3$, both = 10; when $x = 7$, both = 18 — always the same

Equivalent

Write the definition:



$3x + 2x + 5$ simplifies to $5x + 5$ — fewer terms, same value

Simplify

Write the definition:

$$3x + 2x = 5x$$

same variable → combine

$3x$ and $7x$ are like terms (both x); $3x$ and $3y$ are NOT (x vs y); $3x$ and $3x^2$ are NOT (x vs x^2)

Like Terms

Write the definition:

$$2 + 3 = 5$$

$$2 + 3 = 5$$

$4n + 3n = 7n$ — add the coefficients ($4 + 3 = 7$) and keep the variable (n)

Combine

Write the definition:


$$4x$$

coefficient = 4

In $5x + 3$, the coefficient of x is 5 — when $x = 2$, the $5x$ part equals 10

Coefficient

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can show that two expressions are equivalent by simplifying and combining like terms.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equivalent

Simplify

Like Terms

Combine

Coefficient



Tap any word to see what it means and a picture.

1

— Expressions that always have the same value.

2

To write an expression in its shortest equal form is to it.

3

Terms with the same variable raised to the same power, like $4x$ and $9x$, are

.

4

To add or subtract like terms into one term is to them.

5

A number multiplied by a variable is a .



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Which expression is equivalent to $4x + 3x$?

- A. $7x$
- B. $12x$
- C. $7x^2$
- D. $43x$

Step 1 $4x + 3x = 7x$.

Step 2 Combine the coefficients: $4 + 3 = 7$.

 **Answer:** A. $7x$

 Try — put the steps in order

Drag the cards (or use the ▲ ▼ buttons) to put the solution steps in the right order, then press **Check**.

Combine the coefficients: $4 + 3 = 7$.

$4x + 3x = 7x$.

 We do — together

Solve this one with your class using the same steps.

Problem: Which expression is equivalent to $2(m + 5)$?

- A. $2m + 10$
- B. $2m + 5$
- C. $m + 10$
- D. $7m$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which property is shown: $3(x + 7) = 3x + 21$?

- A. Distributive
- B. Commutative
- C. Associative
- D. Identity

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. A track stacks $3x$ and then one more x . Which expression is equivalent to $3x + x$?

- A. $4x$
- B. $3x^2$
- C. 4
- D. $3x$

Show your work:

2. Producer B flipped a formula to $5 + 2x$. Which expression is equivalent?

- A. $2x + 5$
- B. $7x$
- C. $10x$
- D. $5x + 2$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A student says $2x + 3$ and $5x$ are equivalent because both equal 5 when $x = 1$. Is the student correct? Explain why testing one value is not enough to prove equivalence, and find a value of x where they give different results.

Sentence starter: The student is ____ because _____. When $x = 1$: $2(1) + 3 = \underline{\hspace{1cm}}$ and $5(1) = \underline{\hspace{1cm}}$. But when $x = \underline{\hspace{1cm}}$: $2(\underline{\hspace{1cm}}) + 3 = \underline{\hspace{1cm}}$ and $5(\underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$. To prove equivalence, expressions must be equal for ____ values of x .

Show your work:

Reflect — Exit Ticket

Which expression is equivalent to $6n + 4 + 3n - 1$?

- A. $9n + 3$
- B. $9n + 5$
- C. $63n$
- D. $18n + 4$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Equivalent
2. **Notes 2:** Simplify
3. **Notes 3:** Like Terms
4. **Notes 4:** Combine
5. **Notes 5:** Coefficient
6. **We Try (notes):** A. $2m + 10 - 2(m + 5) = 2m + 10$ using the distributive property.
7. **You Try (notes):** A. Distributive — *Multiplying 3 by each term inside the parentheses is the distributive property.*
8. **Try It 1:** A. $4x - 3x + x$ is 3 groups of x plus 1 more group of $x = 4x$. Add the coefficients: $3 + 1 = 4$. Check $x = 3$: $3(3) + 3 = 12$ and $4(3) = 12$.
9. **Try It 2:** A. $2x + 5$ — *Order of addition does not change the value (commutative property): $5 + 2x = 2x + 5$. Check $x = 10$: $5 + 2(10) = 25$ and $2(10) + 5 = 25$.*
10. **Exit Ticket:** A. $9n + 3 - 6n + 3n = 9n$ and $4 - 1 = 3$, so the simplified expression is $9n + 3$.